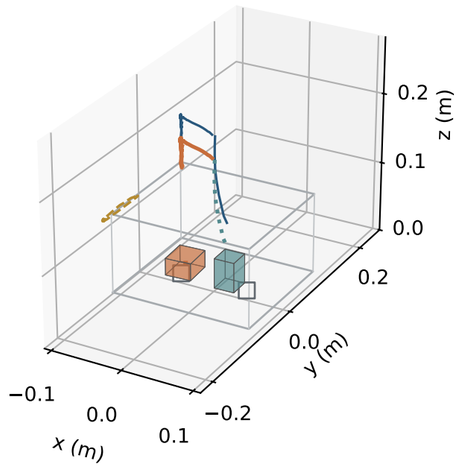
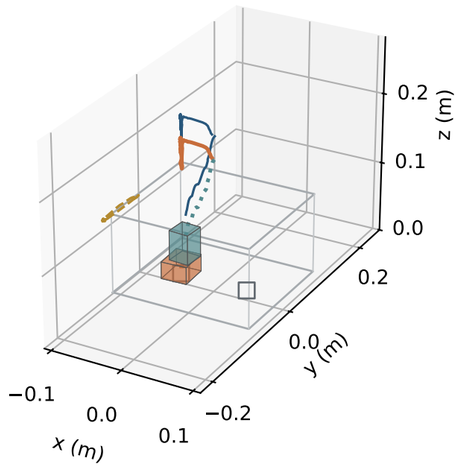
Stage C.1 | Dynamic elastic-collar bag packing with internal occlusion

Left gripper · Right gripper · Item A · Item B

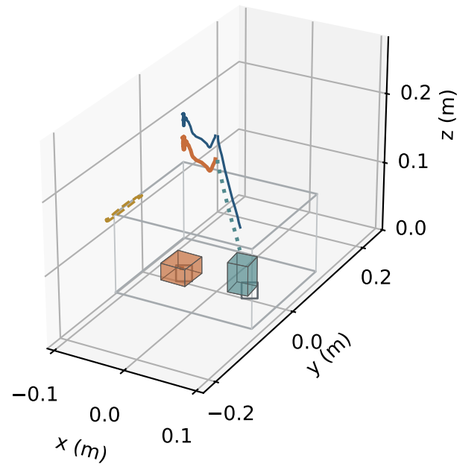
BC (H=4)
Packing: success | Collision: yes



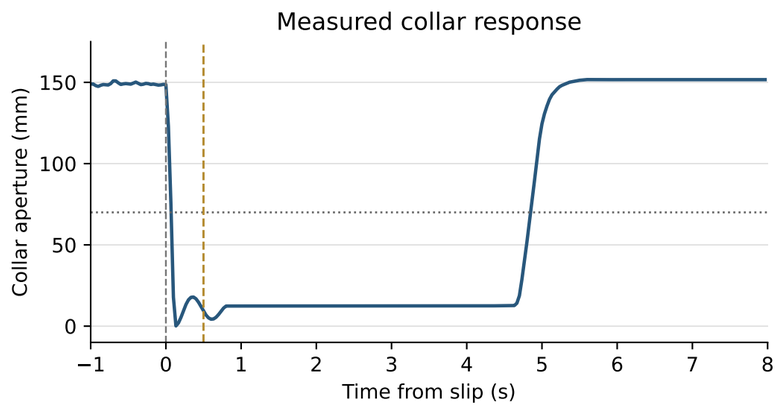
GPC-OPT (M=20)
Packing: failure | Collision: yes



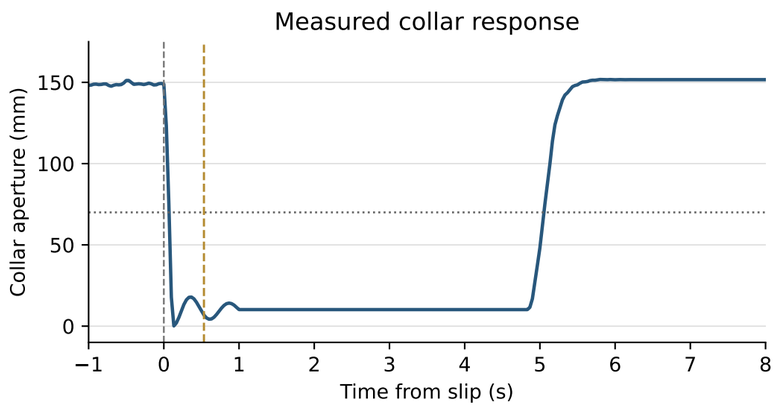
Ours (256D, tau=0.9)
Packing: success | Collision: no



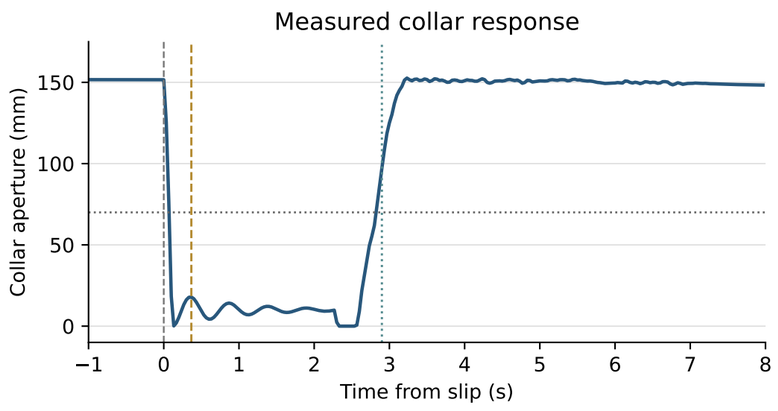
Measured collar response



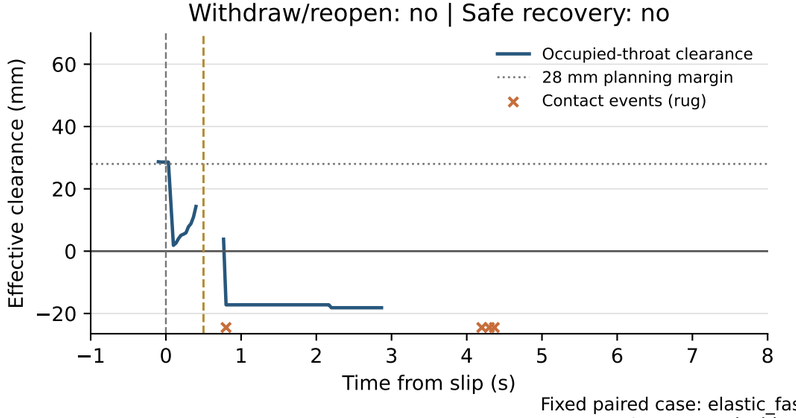
Measured collar response



Measured collar response

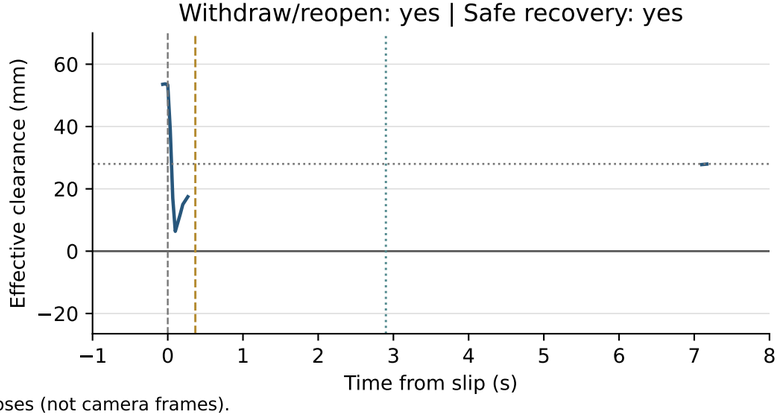


Withdraw/reopen: no | Safe recovery: no



Withdraw/reopen: no | Safe recovery: no

Withdraw/reopen: yes | Safe recovery: yes



Fixed paired case: elastic_fast, A-left then B-right, seed 2000, slip enabled. Top: measured learned-control trajectories and final poses (not camera frames).
Lower panels: identical 9 s event window. Clearance shown only while occupied; orange contact marks are a rug, not clearance values.
Dashed gold: measured withdrawal; dotted teal: measured reopening. Outcomes and unsuccessful recoveries are preserved.
Full-matrix safe recovery | BC H4: 5/18; GPC-OPT: 14/18; Ours tau=0.9: 17/18